

Day 78 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in General Science (5): Kalpakkam's Fast Breeder Reactor & Chandrayaan-2's Lunar Ice Evidence

READING MATERIAL

Covers 2 major 2026 science milestones: India's nuclear energy programme advancing at Kalpakkam, Tamil Nadu, and new lunar water-ice evidence from Chandrayaan-2 data.

Q: What nuclear milestone did India achieve on 6 April 2026, and where?

A: The 500 MWe Prototype Fast Breeder Reactor (PFBR) attained FIRST CRITICALITY (start of its controlled fission chain reaction) at Kalpakkam, Tamil Nadu — India's first indigenous sodium-cooled fast breeder reactor.

Q: Which organisations designed and built the PFBR?

A: Design: Indira Gandhi Centre for Atomic Research (IGCAR). Built & commissioned by: Bharatiya Nabhikiya Vidyut Nigam Limited (BHAVINI).

Q: What fuel does the PFBR use, and what is its core breeding principle?

A: Uranium-Plutonium Mixed Oxide (MOX) fuel, with the core surrounded by a blanket of Uranium-238. Fast neutrons convert the fertile U-238 into fissile Plutonium-239 — meaning the reactor produces MORE fuel than it consumes (hence 'breeder').

Q: What strategic significance does the PFBR's criticality have for India's nuclear programme?

A: It marks India's official entry into the SECOND stage of its three-stage nuclear power programme — a vision first conceived by Dr. Homi Jehangir Bhabha.

Q: What new lunar discovery was reported on 6 May 2026, based on Chandrayaan-2 data?

A: A study (in npj Space Exploration) using radar data from India's Chandrayaan-2 orbiter reported strong evidence for SUBSURFACE WATER ICE in 4 doubly-shadowed craters near the Moon's south pole.

MOCK TEST (WITH ANSWERS)

1. India's Prototype Fast Breeder Reactor (PFBR) attained first criticality on 6 April 2026 at:

- A. Tarapur, Maharashtra
- B. Kalpakkam, Tamil Nadu ✓**
- C. Kudankulam, Tamil Nadu
- D. Kakrapar, Gujarat

Explanation: Kalpakkam, Tamil Nadu. [medium, web-research]

2. The PFBR was designed by which organisation?

- A. BARC (Bhabha Atomic Research Centre)
- B. IGCAR (Indira Gandhi Centre for Atomic Research) ✓**
- C. NPCIL
- D. ISRO

Explanation: Indira Gandhi Centre for Atomic Research (IGCAR). [hard, web-research]

3. The PFBR is called a 'breeder' reactor because:

- A. It runs on solar power as backup
- B. It produces more fuel than it consumes (converting U-238 to Pu-239) ✓**
- C. It breeds new reactor designs
- D. It only uses natural uranium

Explanation: It produces more fuel than it consumes. [medium, web-research]

4. The PFBR's criticality marks India's entry into which stage of its 3-stage nuclear power programme?

- A. First stage
- B. Second stage ✓**
- C. Third stage

D. Final stage

Explanation: Second stage. [hard, web-research]

5. A May 2026 study using radar data from which Indian spacecraft reported strong evidence for subsurface water ice near the Moon's south pole?

A. Chandrayaan-1

B. Chandrayaan-2 ✓

C. Chandrayaan-3

D. Mangalyaan

Explanation: Chandrayaan-2. [medium, web-research]

Part B – Aptitude & Mental Ability: Visual Reasoning

READING MATERIAL

Visual Reasoning is 1 of 6 named Reasoning skills, testing image-based logic rather than text/number puzzles — mirror images, water images, and embedded figures are the standard formats. Note: TNPSC's own answer key has at least one acknowledged disputed mirror-image question, a useful reminder that on visual questions specifically, double-check your own reasoning rather than assuming the 'obvious' answer is automatically right.

Q: What is the rule for a MIRROR image, and what stays the same vs. flips?

A: A mirror image flips LEFT and RIGHT (the right side of the object appears on the left, and vice versa), while TOP and BOTTOM stay the same. Imagine holding the figure up to a vertical mirror.

Q: What is the rule for a WATER image, and how is it different from a mirror image?

A: A water image flips TOP and BOTTOM (it appears upside-down, as if reflected in still water below the object), while LEFT and RIGHT stay the same. This is the opposite axis of flipping compared to a mirror image.

Q: What is the best strategy for solving mirror/water image questions quickly?

A: Focus on the most unique or asymmetric part of the figure first (not the symmetric parts, which look the same either way) — mentally flip just that distinctive part along the correct axis (vertical line for mirror, horizontal line for water), and use it to eliminate options before checking the rest.

Q: What is an 'embedded figure' question, and what does it test?

A: A question where a simple shape is hidden within a more complex, cluttered figure, and you must identify which of the answer-choice shapes appears embedded within it. Tests visual scanning and pattern-matching rather than flipping/rotation.

Q: Real exam (flagged as disputed by TNPSC itself): A figure is shown, and you must choose its correct mirror image from 4 options.

What's the most reliable way to verify your choice, given that even the official answer key has had disputed cases here?

A: Don't just eyeball it — explicitly trace which side each distinctive feature should land on after a left-right flip, point by point, rather than relying on a quick visual impression. Given that TNPSC's own committee has had to review at least one mirror-image answer, treat your worked-through reasoning as more trustworthy than a snap visual judgment, and double-check rather than rushing.

MOCK TEST (WITH ANSWERS)

1. In a MIRROR image of a figure, which pair of sides swap?

- A. Top and bottom
- B. Left and right ✓**
- C. Diagonal corners only
- D. Nothing swaps

Explanation: Left and right swap; top and bottom stay the same. [easy, web-research]

2. In a WATER image of a figure, which pair of sides swap?

- A. Top and bottom ✓**
- B. Left and right
- C. Both pairs swap
- D. Neither swaps

Explanation: Top and bottom swap; left and right stay the same. [easy, web-research]

3. The letter 'b' would appear as which letter in its mirror image (left-right flip)?

- A. b
- B. d ✓**
- C. p
- D. q

Explanation: 'b' has its stem on the left with the bowl attached at bottom-right. Flipping left-right moves the stem to the right with the bowl at bottom-left — that's the letter 'd'. [medium, authored]

4. The letter 'b' would appear as which letter in its water image (top-bottom flip)?

- A. b
- B. d

C. p ✓

D. q

Explanation: Flipping 'b' top-to-bottom keeps the stem on the left but moves the bowl from bottom-right to top-right — that's the letter 'p'. (Mirror and water images of the same letter generally differ — here mirror gives 'd', water gives 'p'.) [hard, authored]

5. An 'embedded figure' question primarily tests:

A. Arithmetic ability

B. Visual scanning and pattern-matching within a cluttered figure ✓

C. Vocabulary

D. Memory of a list

Explanation: Visual scanning/pattern-matching to find a hidden shape within a complex figure. [medium, web-research]

Part C – General English: Grammar: Phrases and Clauses

READING MATERIAL

The key test for this whole topic: a CLAUSE has its own subject and verb; a PHRASE does not. Everything else (noun/adjective/adverb/prepositional phrase or clause; simple/complex sentence) builds on that one distinction.

Q: What is the core difference between a phrase and a clause?

A: A clause contains a subject AND a verb (e.g., 'that he is coming' has subject 'he' + verb 'is coming'). A phrase does NOT contain its own subject (e.g., 'in spite of the rain' has no subject doing an action).

Q: Real exam: identify the phrase in "In spite of the rain, we went out."

A: In spite of the rain — a prepositional phrase (no subject+verb of its own); 'we went out' is the main clause.

Q: Real exam: identify the clause in "I know that he is coming."

A: That he is coming — has its own subject ('he') and verb ('is coming'), making it a (noun) clause, functioning as the object of 'know'.

Q: Real exam: identify the phrase in "On the table lay the keys."

A: On the table — a prepositional phrase telling us WHERE, with no subject+verb of its own.

Q: Real exam: identify the clause in "Because she was tired, she slept early."

A: Because she was tired — has its own subject ('she') and verb ('was'), making it a (adverbial) clause giving the REASON.

Q: Real exam: which of these is a noun phrase — 'with great effort' / 'the tall girl' / 'because he ran' / 'if he comes'?

A: The tall girl — a noun phrase (built around the noun 'girl', no subject+verb of its own); the others are a prepositional phrase and 2 clauses.

Q: Real exam: a dependent clause is also called a _____.

A: Subordinate clause (it depends on/is attached to a main/independent clause and cannot stand alone as a complete sentence).

Q: Real exam: which of these is an adjective clause — 'who was absent' / 'with much anger' / 'jumping quickly' / 'if you go'?

A: Who was absent — an adjective clause (has subject 'who'/verb 'was', and describes/modifies a noun, e.g., 'The student who was absent...').

Q: Real exam: is 'One who does not love one's country is a wretched person' a simple, compound, or complex sentence?

A: Complex — it contains a relative/noun clause ('who does not love one's country') embedded within the main clause.

Q: Real exam: is the phrase 'A crown made of gold' a phrase or a clause, and what type?

A: It is an ADJECTIVE PHRASE (describes the noun 'crown') — not a clause, since 'made of gold' has no subject+verb of its own (it's a reduced/participle phrase, not 'which is made of gold').

Q: Real exam: combine "I saw a Tiger. It was wounded" into one sentence using a phrase/clause-combining technique.

A: I saw a wounded Tiger — the second sentence's information is compressed into a single adjective ('wounded') modifying 'Tiger', avoiding a separate clause.

MOCK TEST (WITH ANSWERS)

1. A clause contains a subject and a _____.

A. Verb ✓

B. Phrase

C. Object

D. Conjunction

Explanation: Verb. [easy, web-research]

2. A phrase does not contain a _____.

A. Subject ✓

B. Modifier

C. Object

D. Preposition

Explanation: Subject. [easy, web-research]

3. Identify the phrase: "In spite of the rain, we went out."

- A. In spite of
- B. In spite of the rain ✓**
- C. We went out
- D. The rain

Explanation: In spite of the rain. [medium, web-research]

4. Identify the clause: "I know that he is coming."

- A. I know
- B. That he is
- C. That he is coming ✓**
- D. He is coming

Explanation: That he is coming. [medium, web-research]

5. Identify the phrase: "On the table lay the keys."

- A. Lay the keys
- B. The keys
- C. On the table ✓**
- D. Table lay

Explanation: On the table. [easy, web-research]

6. Identify the clause: "Because she was tired, she slept early."

- A. She slept
- B. Was tired
- C. She slept early
- D. Because she was tired ✓**

Explanation: Because she was tired. [medium, web-research]

7. Which is a noun phrase?

- A. With great effort
- B. The tall girl ✓**
- C. Because he ran
- D. If he comes

Explanation: The tall girl. [medium, web-research]

8. A dependent clause is also called a _____.

- A. Complete sentence
- B. Subordinate clause ✓**
- C. Phrase
- D. Independent clause

Explanation: Subordinate clause. [easy, web-research]

9. Which is an adjective clause?

- A. Who was absent ✓**
- B. With much anger
- C. Jumping quickly
- D. If you go

Explanation: Who was absent. [medium, web-research]

10. "One who does not love one's country is a wretched person." This is what type of sentence?

- A. Simple
- B. Compound
- C. Complex ✓**
- D. Imperative

Explanation: Complex (contains a relative/noun clause). [medium, question-bank-2024]

11. The phrase "A crown made of gold" is:

- A. An adjective phrase ✓**

- B. An adjective clause
- C. An adverbial phrase
- D. An adverbial clause

Explanation: An adjective phrase. [hard, question-bank-2025]